Pareto Distribution and Power Laws

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Compare with:** Log-normal distribution.

1. Overview

Key point: The **Pareto distribution** (G-1453) models quantities where a few items hold most of the total, such as wealth; it follows a power law, and with the right parameter it gives the famous 80-20 rule.

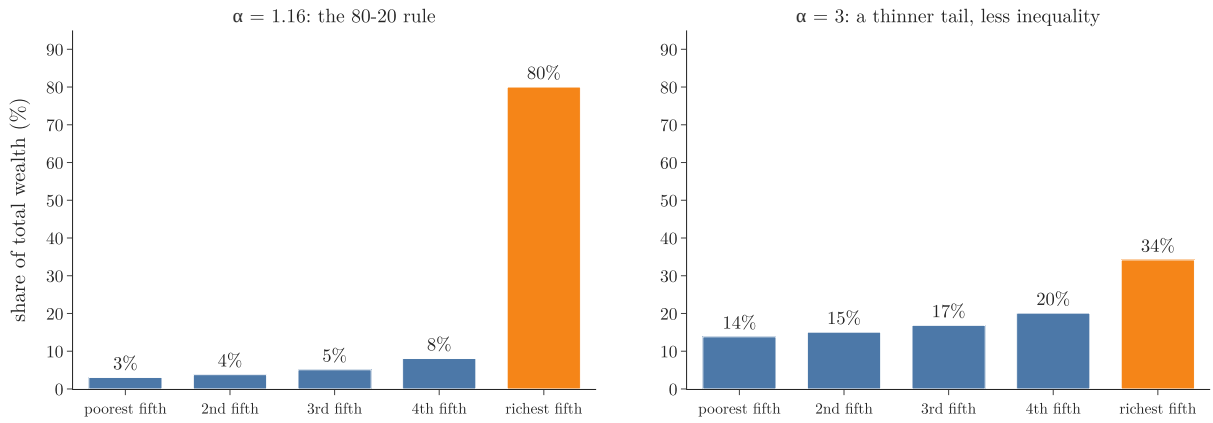


Figure 1: Share of total wealth held by each fifth of the population under a Pareto distribution, for two values of α

Figure 1 shows what the Pareto distribution is known for. On the left, the richest fifth of the population holds 80% of all the wealth and the other four fifths share the remaining 20%. On the right, a different parameter value gives a much more even split.

The Pareto distribution is the third of the famous **non-Gaussian** (G-1334; not shaped like the bell-shaped normal curve) continuous distributions (see Figure 1 of the uniform distribution). The Pareto distribution is common in financial and economic data. This Note covers:

- the **power law** (G-1538) behind it (section 2);
- its PDF and parameters (section 3);
- how to recognise it in data (section 4);
- how non-normal data is made normal (section 5).

2. Power laws

> **Key point:** A power law is a relationship $y = k\,x^a$ in which one variable is proportional to a power of the other.

The **Pareto distribution** (G-1453) is a special case of a **power law** (G-1538). The idea in plain words: every time the first quantity doubles, the second quantity is divided by the same fixed factor, however far along we are.

A small case: let the first quantity be x and the second be y, and suppose each doubling of x divides y by 4. Start at $x = 1$ with $y = 1$.

x	y	what happened
1	1	start
2	0.25	x doubled, y divided by 4
4	0.0625	x doubled again, y divided by 4 again
8	0.015625	the same rule once more

Figure 2 draws these points. On ordinary axes (left) they slide down a curve that starts high and has a long, slowly falling tail. On log-log axes (right), equal steps along an axis mean equal multiplication: on the x axis each step is one doubling, and on the y axis each step is a division by 4. There the points lie on a straight line, because each doubling moves the same distance across and the same distance down. Section 4.1 uses that straight line to recognise a power law in data.

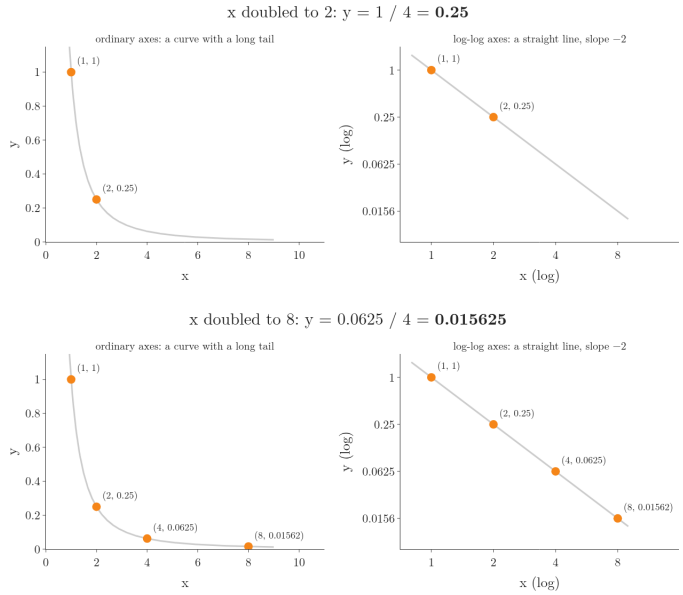


Figure 2: The power law $y = x^{-2}$ at x = 1, 2, 4 and 8: each doubling divides y by 4. Left: ordinary axes. Right: log-log axes, where the points lie on a straight line with slope -2.

The formal version of this rule, with y a fixed power of x, is the power law formula:

$$y = k\,x^a$$

- k is a constant that fixes the starting height. In the table, $k = 1$.
- a is the power. In the table, $a = -2$. A negative power makes y fall as x grows.

Check against the table, one line per row:

$$x = 1: \quad y = 1 \times 1^{-2} = 1$$

$$x = 2: \quad y = 1 \times 2^{-2} = 0.25$$

$$x = 4: \quad y = 1 \times 4^{-2} = 0.0625$$

$$x = 8: \quad y = 1 \times 8^{-2} = 0.015625$$

Doubling x multiplies y by the same factor wherever we start:

$$2^{-2} = \frac{1}{2^2} = \frac{1}{4}$$

so each doubling divides y by 4.

2.1 The 80-20 rule

> **Key point:** The Pareto principle says that roughly 20% of the causes produce 80% of the results, for example 20% of the people hold 80% of the wealth.

The best-known consequence of a **power law** (G-1538) is the **80-20 rule**, or **Pareto principle** (G-54): about 20% of something accounts for about 80% of the result. In wealth, it says that 20% of the population controls 80% of the wealth, and the other 80% of the people share the remaining 20%.

The Italian economist Vilfredo Pareto found this pattern while studying income and wealth data, and published it in his *Cours d'économie politique* (Pareto 1896–97); the distribution is named after him. The 80-20 split is not a law of nature, though: it holds only for one particular value of the distribution's parameter (Section 3.4).

3. The Pareto distribution

> **Key point:** The Pareto distribution is a power-law distribution used to model wealth, income and similar quantities; it starts at a minimum value x_m and its tail is controlled by α.

The **Pareto distribution** (G-1453) is a probability distribution commonly used to model the distribution of wealth, income and other quantities that show power-law behaviour. Its values start at a smallest possible value and fall away in a long right tail, so it is right-skewed (a long tail on the right, see skewness), like the log-normal distribution.

In plain words, picture the savings of a town. Nobody has less than a floor amount, say 1 lakh rupees. Most people hold a little above the floor, and a few hold enormous amounts. The distribution describes how quickly the number of people thins out as the amount grows. Two numbers describe it: the floor, and how fast the thinning happens (section 3.1). When the thinning is slow, a few people hold a huge share of everything; when it is fast, wealth is spread more evenly (section 3.3).

3.1 Parameters

> **Key point:** Two parameters: the minimum value x_m (scale) and the shape α, which sets how fat the tail is.

- x_m**, the minimum value (scale):** the point on the x axis where the curve starts. No value is smaller than x_m.
- α**, the shape (tail index):** a positive number that controls the tail.

The **Pareto distribution** (G-1453) is sometimes described as having one parameter, α, because x_m only sets the starting point and the units. But both are needed to write the PDF.

3.2 The Pareto PDF

> **Key point:** The density is highest at x_m, where it equals α/x_m, and falls as a power of x after that.

Take the small case $x_m = 1$ and $\alpha = 3$. The density at the minimum is the highest, and each doubling of x divides it by $2^4 = 16$. The steps:

x	numerator αx_m^α	denominator $x^{\alpha+1}$	density $f(x)$
1	$3 \times 1^3 = 3$	$1^4 = 1$	3
2	$3 \times 1^3 = 3$	$2^4 = 16$	$3/16 = 0.1875$
4	$3 \times 1^3 = 3$	$4^4 = 256$	$3/256 = 0.0117$

The same recipe for any x at or above the minimum, with the parameters named in section 3.1, is the **Pareto PDF** (probability density function: the curve whose height shows how likely values near x are; see the density at a point):

$$f(x) = \frac{\alpha\, x_m^{\alpha}}{x^{\alpha+1}}$$

The formula holds for $x \geq x_m$.

Below x_m the density is 0. The formula is a **power law** (G-1538) $k\, x^a$ with $k = \alpha x_m^{\alpha}$ and $a = -(\alpha + 1)$; for the table, $k = 3$ and $a = -4$, which matches the rows above.

Figure 3 shows the PDF (left) and the CDF (right) for three values of α; section 3.3 reads it. The **CDF** (cumulative distribution function, see the CDF of a continuous variable) gives, for each x, the share of values at or below x.

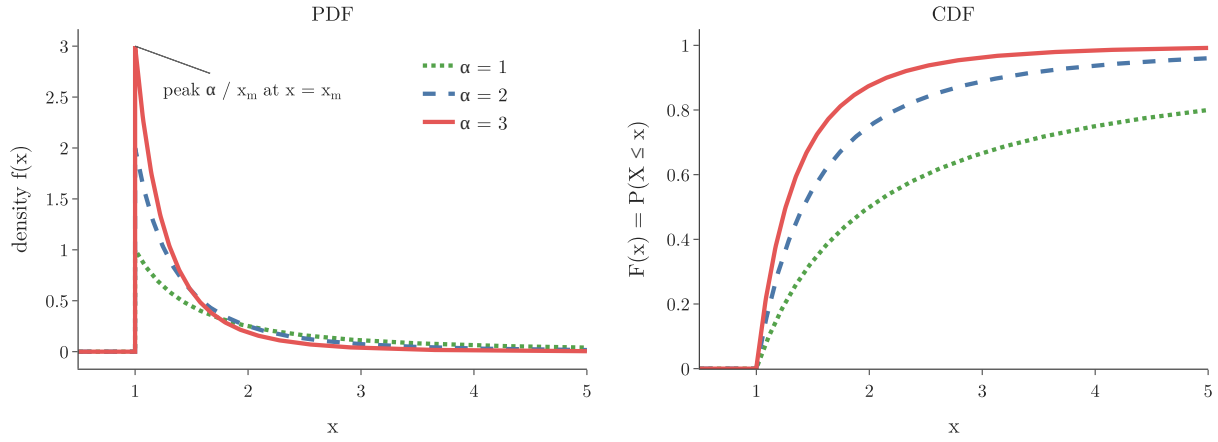


Figure 3: Pareto PDFs (left) and CDFs (right) with $x_m = 1$ and three values of α

3.3 What α does

> **Key point:** A larger α gives a higher peak at x_m and a thinner tail; a smaller α gives a lower peak and a fatter tail.

Figure 3 (left) shows three curves with $x_m = 1$:

- $\alpha = 3$ **(red):** the highest peak, 3, and the thinnest tail. Almost all values are close to x_m.
- $\alpha = 2$ **(blue):** in between.
- $\alpha = 1$ **(green):** the lowest peak, 1, and the fattest tail. Very large values are relatively common.

As α grows without limit, the whole curve collapses into a single vertical spike at x_m: every value equals x_m and there is no tail at all.

In wealth terms, a fat tail (small α) means a few individuals hold huge amounts, so the inequality is **larger**. A thin tail (large α) means most people hold amounts near the minimum and the split is more even, as in Figure 1 (right). Figure 4 turns α from 1.16 to 3 and shows

the share of the total held by each fifth of the population (the formula is in the Extra of section 3.4). The richest fifth's share falls from 80 to 58, 45 and finally 34 percent.

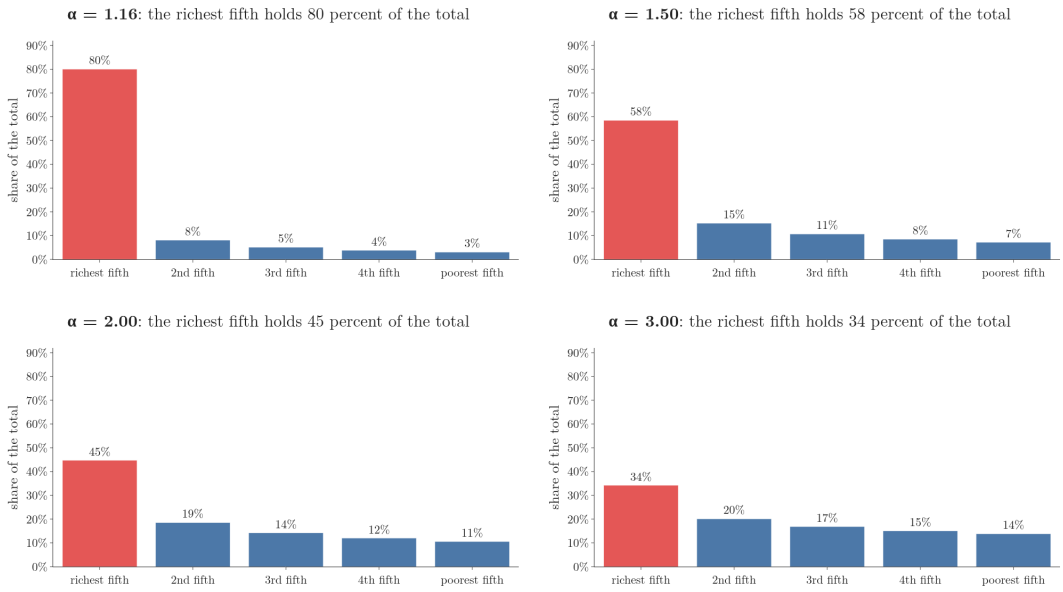


Figure 4: The share of the total held by each fifth of a Pareto population, for $\alpha = 1.16$, 1.5, 2 and 3. A larger α gives a more even split.

3.4 The CDF and the 80-20 rule

> **Key point:** The CDF is $1 - (x_m/x)^\alpha$; the 80-20 split holds only for $\alpha \approx 1.16$.

The idea: pick a value x and ask what share of the population lies above it. For a **Pareto distribution** (G-1453) that share falls by a fixed recipe, and the share at or below x is what remains.

Take $x_m = 1$ and $\alpha = 3$, and look at $x = 2$:

$$\frac{x_m}{x} = \frac{1}{2} = 0.5$$

$$\left(\frac{x_m}{x}\right)^\alpha = 0.5^3 = 0.125 \quad \text{(share above 2)}$$

$$F(2) = 1 - 0.125 = 0.875 \quad \text{(share at or below 2)}$$

With $\alpha = 1$ the same steps give the share above 2:

$$0.5^1 = 0.5$$

Half the population is still above 2: the fat tail.

The formal version is the **CDF** $F(x)$ (the share of values at or below x):

$$F(x) = P(X \leq x)$$

$$F(x) = 1 - \left(\frac{x_m}{x}\right)^{\alpha}$$

The formula holds for $x \geq x_m$.

Here $P(X \leq x)$ reads "the probability that a randomly chosen value X is at most x". For $x = 2$, $\alpha = 3$, it gives the 0.875 above.

In Figure 3 (right), a smaller α makes the CDF climb to 1 more slowly. The slow climb is the fat tail again: a sizeable share of the values lies far out, so it takes a long way along the x axis to collect all of them. A large α reaches 1 quickly, because almost everything sits just above x_m.

Extra: The share of the total held by the richest fraction p of a Pareto population is $p^{1-1/\alpha}$ (for $\alpha > 1$).
The idea: the richest 20% are everyone above some cut-off x_p, and we compare what they hold with what everyone holds. Take $x_m = 1$ and $\alpha = 3$, top $p = 0.2$:

$$\left(\frac{1}{x_p}\right)^3 = 0.2$$

$$x_p = 0.2^{-1/3}$$

$$x_p = 1.710$$

$$\text{share held} = \left(\frac{x_m}{x_p}\right)^{\alpha-1}$$

$$= \left(\frac{1}{1.710}\right)^2$$

$$= 0.342$$

So the top 20% hold 34% (Figure 1, right). For general α the same steps, in symbols:

1. The richest fraction p are the values above x_p, where $(x_m/x_p)^{\alpha} = p$.
2. Their total is:

$$\int_{x_p}^{\infty} x f(x)\, dx$$

$$= \frac{\alpha x_m^{\alpha}}{\alpha - 1} x_p^{1-\alpha}$$

The symbol $\int_{x_p}^{\infty}$ means "add up $xf(x)$ over every value from x_p upward".

3. The total of everyone is the same integral from x_m:

$$\frac{\alpha x_m}{\alpha - 1}$$

4. Dividing step 2 by step 3 gives the share:

$$\left(\frac{x_m}{x_p}\right)^{\alpha-1}$$

$$= p^{(\alpha-1)/\alpha}$$

The result is the share formula, with p the fraction of the population (0.2 means 20%):

$$\text{share held by the top } p = p^{1-1/\alpha}$$

Check against the **80-20 rule** (G-54), with $\alpha = 1.16$:

$$1 - 1/1.16 = 0.139$$

$$0.2^{0.139} = 0.80$$

and with $\alpha = 3$:

$$1 - \frac{1}{3} = \frac{2}{3}$$

$$0.2^{2/3} = 0.34$$

This matches the lines above.
The exact value for the 80-20 rule is $\alpha = \log_4 5 = 1.161$. Also, the mean of a Pareto distribution, $\alpha x_m/(\alpha - 1)$, exists only for $\alpha > 1$: with $\alpha \leq 1$ the tail is so fat that the average is infinite.

3.5 Where the Pareto distribution appears

Key point: Wealth and income at the top, city and settlement sizes, and file sizes on the internet.

- **Wealth and income:** a small share of people holds most of the wealth. The Pareto tail describes the richest few percent; the rest of the incomes are closer to log-normal (see where the log-normal appears).
- **Human settlements:** many people crowd into a small share of the land, and a few live spread over remote areas. Very roughly, 20% of the area holds 80% of the population.
- **File sizes in internet traffic:** most files are small and a few are huge, so a small

share of the files makes up most of the gigabytes transferred. Measured web file sizes follow a heavy, Pareto-like tail (Crovella and Bestavros 1997).

Python: The **Pareto distribution** (G-1453) in scipy.

```
from scipy import stats

# b = alpha, scale = x_m (default 1)
p = stats.pareto(b=3, scale=1)
p.pdf(2)        # 0.1875
p.sf(2)         # 0.125, P(X > 2)
data = p.rvs(size=1000, random_state=42)
```

The skewness of a Pareto distribution has a formula, but in practice we let the library compute it from data with `stats.skew`.

4. How to check whether data is Pareto

Key point: Two checks: a log-log plot, where a power law becomes a straight line, and a Q-Q plot against a fitted **Pareto distribution** (G-1453).

"Given a distribution, how would we know that it is Pareto?" is another common interview question. There are two answers.

4.1 The log-log plot

Key point: Taking logs of both sides turns a **power law** (G-1538) into a straight line, with the power as its slope.

The idea: a power law falls by a fixed factor at each doubling, so after taking the log of both quantities each doubling is the same step. Plot $\ln y$ against $\ln x$ (ln is the natural logarithm; this is a **log-log plot**, G-1114) and the points lie on a straight line.

Take the Pareto PDF with $x_m = 1$ and $\alpha = 3$, which is $f(x) = 3/x^4$. Three points:

x	$\ln x$	$f(x)$	$\ln f(x)$
1	0	3	1.099
$e = 2.718$	1	$3/e^4 = 0.0549$	-2.90
$e^2 = 7.389$	2	$3/e^8 = 0.00101$	-6.90

Each step of 1 in $\ln x$ lowers $\ln f$ by 4.0: a straight line of slope -4 (Figure 5, left). The general formula behind it, for any x_m and α, is:

$$\ln f(x) = \ln(\alpha x_m^\alpha) - (\alpha + 1) \ln x$$

A constant minus $(\alpha + 1)$ times $\ln x$ is a line with slope $-(\alpha + 1)$. For the table, the constant and the slope are:

$$\text{constant} = \ln 3 = 1.099$$

$$\text{slope} = -(3+1) = -4$$

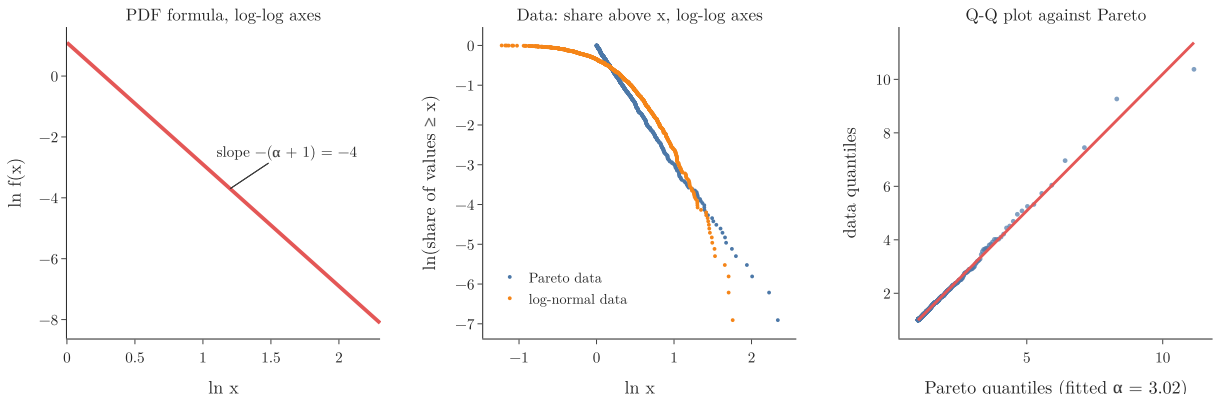


Figure 5: Left: the Pareto PDF on log-log axes. Middle: data on log-log axes, Pareto against log-normal. Right: Q-Q plot against a fitted Pareto

Plotting the PDF formula itself always gives a straight line, so Figure 5 (left) only shows what to look for. With real data we do not know the PDF. A practical version plots, for every data value x, the share of values at or above x. For Pareto data that share is:

$$\left(\frac{x_m}{x}\right)^{\alpha}$$

Its log is

$$\alpha \ln x_m - \alpha \ln x$$

again a straight line, now with slope $-\alpha$.

Figure 5 (middle) does this for 1,000 values from a **Pareto distribution** (G-1453) ($\alpha = 3$, blue) and 1,000 values from a log-normal distribution (orange). Both are right-skewed, but only the Pareto values give a straight line; the log-normal ones bend downwards. The scattered points at the far right are the few largest values; the next paragraph measures how much they vary.

4.2 The Q-Q plot against a Pareto distribution

> **Key point:** Fit a **Pareto distribution** (G-1453) to the data, then draw a Q-Q plot against it; points on the line mean Pareto.

A **Q-Q plot** compares the quantiles (cut points) of the data with those of a distribution; points on a straight line mean a match. It can be drawn for any distribution (see building a Q-Q plot). Here the theoretical distribution is a Pareto whose α is estimated from the data itself.

Figure 5 (right) shows the 1,000 Pareto values against a fitted Pareto ($\alpha = 3.02$, close to the true 3). Most points lie on the line; the last few, the largest values, move off it a little. The Notebook measures why: over 500 fresh samples of 1,000 Pareto values ($\alpha = 3$), the middle (500th) value has a spread (standard deviation over mean) of about 1%, while the largest value has a spread of about 55%. The biggest values of a heavy tail vary a lot from sample to sample, so a few points off the line at the end are expected. So the data is roughly Pareto.

> **Python:** Fit, then Q-Q plot.
>
> ```
> # estimate alpha with the start fixed at 0 (loc)
> b, loc, scale = stats.pareto.fit(data, floc=0)
> (osm, osr), (slope, icpt, r) = stats.probplot(
>     data, dist=stats.pareto, sparams=(b, loc, scale))
> ```
>
> `sparams` passes the fitted parameters to the theoretical distribution. The Notebook (`MA-030-pareto-and-power-law.ipynb`) draws all three panels of Figure 5 with Plotly.

5. Making data normal: transformations

> **Key point:** Real data is rarely normal; mathematical transformations such as the log, square root, reciprocal, Box-Cox and Yeo-Johnson bring it close to normal.

The normal distribution is so well studied that once data is normal, many calculations and methods become easy. Statistical models such as linear and logistic regression work better when each **feature** (G-772; an input variable, one column of the data table) looks normal, while tree-based models do not care. A tree only asks questions such as "is the value above 3?", and a transform like the log keeps the values in the same order, so every question splits the observations exactly as before (scikit-learn user guide §1.10: trees need little data preparation). Real data, however, is rarely normal; it is often skewed like the log-normal and Pareto data above.

Mathematical transformations (G-1174; see mathematical transformers in scikit-learn) apply a formula to every value of a feature to bring its distribution close to normal. They are covered in the feature engineering Notes:

Transformation	What it does	Taught in
Log, $\log(1+x)$	pulls in a long right tail; makes log-normal data normal	function transformer
Reciprocal, square, square root	other fixed formulas, for other shapes of skew	function transformer
Box-Cox	learns the best power λ; positive values only	**power transformer** (G-1542)
Yeo-Johnson (G-2136)	Box-Cox adapted to zeros and negative values	**power transformer** (G-1542)

Each of those Notes checks the result with a Q-Q plot and shows the effect on a model's score, so the workflow is:

1. check the feature (density plot, skewness, Q-Q plot);
2. transform it;
3. check again.

Figure 6 shows why step 3 matters, on the 1,000 Pareto values of section 3.5. The log cuts the skewness from 7.68 to 1.87. Box-Cox picks the power $\lambda = -2.17$ and cuts it to 0.30, yet its histogram is still flat-topped, not a bell. A skewness near 0 alone does not prove normality (see symmetric does not mean normal); the second check would catch this.

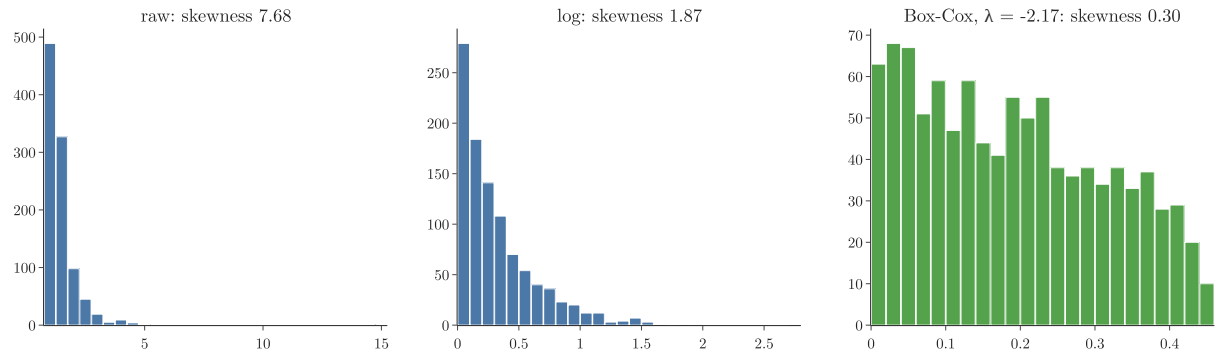


Figure 6: 1,000 Pareto values ($\alpha = 3$): raw, after the log, and after Box-Cox. The skewness falls from 7.68 to 1.87 to 0.30, but the Box-Cox histogram is still not a bell.

6. Summary

Idea	Formula	Example ($x_m = 1$, $\alpha = 3$)	Why it matters
Power law	$y = k\,x^a$	$k = 1$, $a = -2$: $y(2) = 0.25$	the rule behind the Pareto distribution
Pareto PDF	$\alpha x_m^\alpha / x^{\alpha+1}$, $x \geq x_m$	$f(2) = 0.1875$	highest at x_m, then falls as a power of x, so most values are small and a few are huge
Pareto CDF	$1 - (x_m/x)^\alpha$	$F(2) = 0.875$	gives the share at or below x directly, so no table is needed
Share held by top p	$p^{1-1/\alpha}$	top 20%: 34% ($\alpha = 1.16$: 80%)	says how unequal the split is for a given α
Log-log PDF	slope $-(\alpha + 1)$	slope -4	a straight line on log-log axes is the sign of a power law

- Larger α: higher peak at x_m, thinner tail, less inequality. Smaller α: fatter tail, more inequality, because a fat tail means a few items hold huge amounts.
- The **80-20 rule** (G-54) holds only for $\alpha \approx 1.16$, so "Pareto" does not by itself mean an 80-20 split ($\alpha = 3$ gives 34%).
- Check for Pareto with a log-log plot (straight line) or a Q-Q plot against a fitted Pareto, because right skew alone does not separate Pareto from log-normal data, whose log-log line bends.
- Non-normal features are brought close to normal with the transformations of the function and **power transformer** (G-1542) Notes, because models such as linear and logistic regression work better on normal-looking features.
- This answers the opening question: the **Pareto distribution** (G-1453) follows a power law in which a few items hold most of the total, and $\alpha \approx 1.16$ gives the 80-20 rule.

7. Sources

Built from

- CampusX, "Session 42 - Non-Gaussian Probability Distributions | DSMP 2023", YouTube, https://www.youtube.com/watch?v=U6QCc_3zgUk

Other references

- Pareto, V. (1896–97). *Cours d'économie politique*, 2 vols. Lausanne: F. Rouge.
- scikit-learn user guide, §1.10 "Decision Trees", advantages (requires little data preparation). https://scikit-learn.org/stable/modules/tree.html

- Crovella, M. E. and Bestavros, A. (1997). "Self-similarity in World Wide Web traffic: evidence and possible causes." *IEEE/ACM Transactions on Networking* 5(6), 835–846.

8. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Pareto distribution (G-1453)	A distribution of values that start at a smallest value x_m and become rarer the larger they are, in a long right tail (a right-skewed power law); used to model wealth and income. PDF $\alpha x_m^\alpha / x^{\alpha+1}$.
Power law (G-1538)	A relationship $y = k\,x^a$: one variable proportional to a power of the other.
80-20 rule (Pareto principle) (G-54)	About 20% of the causes produce about 80% of the results, e.g. 20% of people hold 80% of the wealth.
Log-log plot (G-1114)	A plot with both axes on a log scale: it plots $\ln y$ against $\ln x$. On it a power law shows as a straight line.
α (Pareto shape) (G-2)	The number that sets how fat the tail of a Pareto distribution is (its shape parameter, also called the tail index). A larger α gives a thinner tail and fewer extreme values.
x_m (Pareto) (G-43)	The minimum possible value, where the Pareto curve starts and peaks.
Non-Gaussian distribution (G-1334)	Any distribution that is not normal.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Mathematical transformation (G-1174)	Applying one mathematical formula, such as the log or the square root, to every value of a column, usually to bring its distribution closer to normal.
Power transformer (G-1542)	A transform that raises each column to a learned power λ to make it close to normal.
Yeo-Johnson transform (G-2136)	A power transform that makes a column closer to normal, like Box-Cox but also working on zero and negative values; scikit-learn's default.